Rohan Siva

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Education

University of Texas at Austin — Bachelor of Science, Electrical & Computer Engineering (Honors)
GPA: 3.98/4.00 — *Expected May 2027*

Experience

UT Austin Center for Autonomy / VITA Lab — August 2024 – Present
Research Lead — *Austin, Texas*

- Led first-author **CoRL 2026** research on functionality-based environment understanding for robotic manipulation; learned transferable object affordances, achieving **94% accuracy** and **100× faster inference**.
- Developed LSTM probes on VLA latent states to detect manipulation failure modes ahead of time for corrective steering; achieved **95% failure recall** on unseen tasks.
- Fine-tuned GR00T N1.7 and $\pi_{0.5}$ action policies via imitation learning on teleoperated data; used DeepSpeed ZeRO-2/NCCL to shard optimizer states and gradients under limited compute; raised task success **7% to 83%**.
- Built a zero-shot vision-language navigation planner, using ROS/SLAM to build scene graphs and caching trajectories; achieved **2× success** and **55% fewer VLM calls** with sim-to-real transfer on Spot.
- Used conformal prediction to filter LLaVA training data; cut training GPU memory by **70%** with **LoRA** adapters.

Decagon — June 2026 – August 2026
Research Intern — *San Francisco, California*

- Co-developed **Chord, Decagon's first in-house speech model**, enabling fine-grained speech tags; improved voice realism through custom rewards and data filtering, with **90%+ human-indistinguishability**.
- Designed reinforcement learning algorithms from scratch for autoregressive/flow-matching TTS: **Flow-DPO** using loss substitution for preference optimization and **Flow-GRPO** using stochastic ODE-to-SDE policy gradients.
- Pioneered Decagon's **production TTS inference engine**, building custom model inference from scratch on M*; reduced serving GPUs by **over 5×** with stage-aware scheduling, packed prefill, and paged KV caching.
- Independently batched AR/flow generation and causal audio decoding with per-request history and CUDA Graphs; achieved **27–30 ms** first-audio latency (**3× faster**) and **10× throughput** vs. vLLM-Omni at high concurrency.

Mercor — Aug 2025 – Dec 2025
Machine Learning Engineer Intern — *Remote*

- Built a RAG evaluation pipeline for **50+** OpenAI MLE tasks daily, reducing manual review time by **85%**
- Optimized LLM evaluation with embedding caching, achieving **sub-200 ms** latency and **30%** lower compute costs

Cisco Systems — May 2025 – Aug 2025
Machine Learning Engineer Intern — *San Francisco, CA*

- Developed **kRAIG**, a natural-language data pipelining agent that generates Kubeflow ELT workflows via tool retrieval and intent clarification; adopted by **9+ teams**, cutting pipeline tickets **80%** and setup time **70%**.
- Designed **ReQuesAct (Reason–Question–Act)**, boosting 100-task ELT-Bench extract/load success **74% to 91%**.
- Deployed kRAIG via React, Elasticsearch, AWS Bedrock, and Lambda microservices, maintaining **99.7%** uptime

Statistical Learning & AI Group Lab @ UT Austin — May 2025 – Aug 2025
Researcher — *Remote*

- Accelerated GRPO rollouts with trajectory-conditioned EAGLE **speculative decoding**; raised acceptance by **15%**
- Implemented GRPO/PPO variants for NanoChat, improving post-RL GSM8K performance by about **12%**

PKU Lab @ Peking University — May 2025 – Aug 2025
AI Researcher — *Remote*

- Studied block-by-block diffusion for uncertainty-aware multimodal CoT planning, advised by Prof. Hao Tang.
- Designed parallel diffusion reasoning with real-time posterior variance feedback; reduced CoT latency by **34%** for VLM-based autonomous driving.

Technical Skills

Robotics: ROS/ROS 2, SLAM, Isaac Sim/Isaac Lab, MuJoCo, PhysX, Newton, CARLA, Habitat, LIBERO/LIBERO-X
ML & Sys: PyTorch, CUDA, Triton, JAX, Transformers, CLIP, OpenCV, DeepSpeed, FSDP, Docker, Kubernetes, AWS
Languages: Python, Java, C/C++, JavaScript, SQL, TypeScript, HTML/CSS

Selected Publications

Siva, R., Bhatt, N. P., Yang, Y., Lee, S., Gadde, N., Ellis, C., Velasquez, A., Wang, Z., & Topcu, U. "What Objects Enable, Not What They Are: Functional Latent Spaces for Affordance Reasoning," In Proceedings, **Conference on Robot Learning (CoRL), 2026**.

Siva, R., Cheung, K., Li, L., & Sundaram, G. "kRAIG: A Natural Language-Driven Agent for Automated DataOps Pipeline Generation," In Proceedings, **Conference on Neural Information Processing Systems (NeurIPS), 2026**.

Bhatt, N. P., Yang, Y., **Siva, R.**, Samineni, P., Milan, D., Wang, Z., & Topcu, U. "VLN-Zero: Rapid Exploration and Cache-Enabled Neurosymbolic Vision-Language Planning for Zero-Shot Transfer in Robot Navigation," In Proceedings, **NeuS, 2026. (Spotlight)**

Bhatt, N. P., Yang, Y., **Siva, R.**, Milan, D., Topcu, U., & Wang, Z. "Know Where You're Uncertain When Planning with Multimodal Foundation Models: A Formal Framework," In Proceedings, **Conference on Machine Learning and Systems (MLSys), 2025. (Oral)**

Bhatt, N. P., Li, P., Gupta, K., **Siva, R.**, Milan, D., Hogue, A. T., Chinchali, S. P., Fridovich-Keil, D., Wang, Z., & Topcu, U. "UNCAP: Uncertainty-Guided Planning Using Natural Language Communication for Cooperative Autonomous Vehicles," In Proceedings, **International Conference on Autonomous Agents and Multiagent Systems (AAMAS), 2026. (Oral + Best Paper Finalist)**

Projects

Click — From-Scratch Computer-Use Vision-Language Model & RL Stack *2026*

- Built a **custom Tensor/autograd engine** with reverse-mode differentiation, dynamic graphs, broadcast-aware gradients, and Adam; trained a tiny transformer from scratch.
- Pretrained a **300M-parameter computer-use VLM** (ViT, GQA, RoPE, SwiGLU) on **4× A6000s**; SFT and GRPO/GSPO (group advantages, clipping, KL) on synthetic Playwright GUIs raised success **48% to 72%**.
- Built batched RL rollouts with shared-prefix KV caching and custom Triton/CUDA kernels for RMSNorm, RoPE, log-probs, and fused losses; achieved **2.5× speedup** vs. PyTorch eager and **20% faster RL steps**.

Where Did the Sink Go? — Diffusion Transformer Interpretability *2026 | Research Blog*

- Collaborated with a Google DeepMind researcher to find LLM attention sinks and massive activations in multimodal DiTs (SD3.5, Qwen-Image, HunyuanVideo); sinks migrate to text at **294–359×** uniform attention.
- Traced attention, activation norms, and adaLN gates; Qwen-Image text tokens received **21%** of attention despite comprising **0.3%** of tokens, with a dominant image channel in **59/60 layers**.

Robot Safety Simulation & Failure-Data Benchmark *2026*

- Built a **UR10e robot safety simulator** in Isaac Sim/Lab and MuJoCo for handovers and human interaction; cross-validated separation/contact metrics across PhysX and Newton/MuJoCo-Warp.
- Exported **184 LeRobot v3 episodes** with **8 fault types**, **4 human behaviors**, and failure labels.
- In **160 paired trials**, safety controls cut human contacts **81%** but raised deadlocks to **46%** and reduced task completion **78% to 50%**.